

CARBON OFFSETS

Tropical forest carbon offsets deliver partial gains amid persistent over-crediting

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REDD+ (Reducing Emissions from Deforestation and Degradation Plus) projects generate carbon credits to offset emissions, but recent studies have questioned their effectiveness. We evaluated 52 voluntary REDD+ projects across 12 tropical countries using synthetic control methods. Only a minority of project units showed statistically significant reductions in deforestation, and just 19% met their reported emissions targets. Nonetheless, many underperforming projects still delivered partial climate benefits, with an estimated 13.2% of tradable credits supported by counterfactual analysis. Effectiveness varied by region, with stronger performance in Brazil and Africa. Although systematic over-crediting remains a concern, our results suggest greater climate benefits than previous assessments. Improving baseline methodologies and strengthening verification frameworks will be essential for enhancing the credibility and impact of forest carbon offsets.

Tropical forests play a crucial role in supporting biodiversity and ecosystem services (1–3), and their conservation is critical for sustainable development and nature-based climate solutions (4). Unfortunately, forest loss and degradation is intense (5–7), leading to the emission of ~5 billion metric tons of carbon dioxide annually—second only to fossil fuel use (8). The REDD+ (Reducing Emissions from Deforestation and Degradation Plus) program is an international initiative aimed at reducing greenhouse gas emissions by offering economic incentives to developing countries to maintain forest cover, enhance carbon stocks, and promote human well-being (9, 10).

There are currently more than 350 voluntary REDD+ projects operating worldwide (11), many of which are funded through the sale of carbon offsets. After their initial development, REDD+ credits gained considerable popularity; by 2021, they accounted for two-thirds of the 227.7 million offsets traded in carbon markets from the land-use sector (excluding agriculture), with a total market value of \$1.3 billion (12). However, since their inception, the efficacy of REDD+ projects has been a subject of debate (13–16). Over time, this scrutiny has led to a series of quantitative studies that have challenged the claimed climate benefits of many projects (17–20). These critiques have shaken confidence in the voluntary carbon market (21, 22), leading to a 62% drop

in the value of REDD+ carbon offsets between 2022 and 2023 (23). To restore trust and accurately assess the climate benefits of REDD+ projects, there is an urgent need for more robust evaluation methodologies that incorporate adequate controls, sufficient sample sizes, and transparent analytical approaches (24–29).

In this work, we assessed the climate change mitigation potential of voluntary REDD+ projects certified by Verra's Verified Carbon Standard (VCS) across several tropical regions worldwide. We focused on Avoided Unplanned Deforestation (AUD) projects, which are a crucial component of REDD+ efforts in many developing countries (15, 30), that aim to protect forests from localized deforestation driven by small-scale land uses, such as subsistence agriculture, illegal logging, and fuelwood collection (31–33). We screened VCS AUD projects (34) for those that provided the necessary geospatial and landscape data to apply forest loss baseline comparison methods based on synthetic control models [see (35); table S1 and table S2]. In total, we analyzed 54 REDD+ projects across half of the countries where such projects are implemented, including those in South America (Brazil, Colombia, and Peru), Africa (Central African Republic, Democratic Republic of Congo, Madagascar, Mozambique, Sierra Leone, Tanzania, and Zambia), and Southeast Asia (Cambodia and Laos) (Fig. 1). Some projects encompassed geographically dispersed sites, which we treated as separate project units, resulting in 69 distinct project units for our analyses [(35); fig. S1]. Although we could not analyze several projects that were included in prior evaluations owing to the inapplicable data or undisclosed results, we did incorporate data from many (31) of the same projects previously assessed (17, 20) as well as from many projects that had not previously been subjected to synthetic analyses (35).

We quantified the climate mitigation effectiveness of REDD+ projects at both individual project and regional scales. To do so, we estimated the potential impact of each REDD+ project by constructing a synthetic counterfactual using a weighted average of nearby synthetic control areas selected to match the project's preintervention deforestation trends and key biophysical and socioeconomic conditions [(35); fig. S2 and table S3]. In the absence of intervention, the project area should exhibit a similar level of forest loss to its synthetic controls (36, 37). We selected control areas within the REDD+ project buffer zone (table S4), excluding nonnational territory and leakage buffers (figs. S3 to S6), and chose a majority of time-variant key geospatial predictors of deforestation risk for control matching (35). For within-study analyses, we generated multiple donor pools after similarity screening using Earth Mover's Distance (38), selecting the most representative subsets of control areas (35). To establish the forest loss counterfactual for each sample pool, we used the penalized synthetic control method (PSCM), which balances pairwise and overall matching discrepancies (39). We applied the PSCM to annual forest loss to match preintervention annual trends. We eliminated project units (1953-1, 1953-2, and 1748) that performed poorly in the model fitting and validation test to ensure adequate test causality (giving us a total sample of 66 units within 52 projects). Next, we evaluated average cross-project impacts when data were aggregated at larger scales using the generalized synthetic control method (GSCM), which incorporates interactive fixed effects to account for preexisting differences and trends between treatment and control groups (33). We applied this approach to countries or regions with sufficient project units: (i) each of three countries in South America, Brazil ($N = 28$), Colombia ($N = 12$), and Peru ($N = 13$); (ii) regional or continental scales in Africa ($N = 10$) and Southeast Asia ($N = 3$); and (iii) all pantropical sites combined ($N = 66$). We cross-validated these results with synthetic difference-in-differences (SDID) modeling, which evaluates treatment effects by optimizing the weights of control units over time. Lastly, we estimated the potential over-crediting of REDD+ projects by contrasting forest loss baselines derived from synthetic controls with those reported in project documents, which is crucial for assessing carbon credit reliability.

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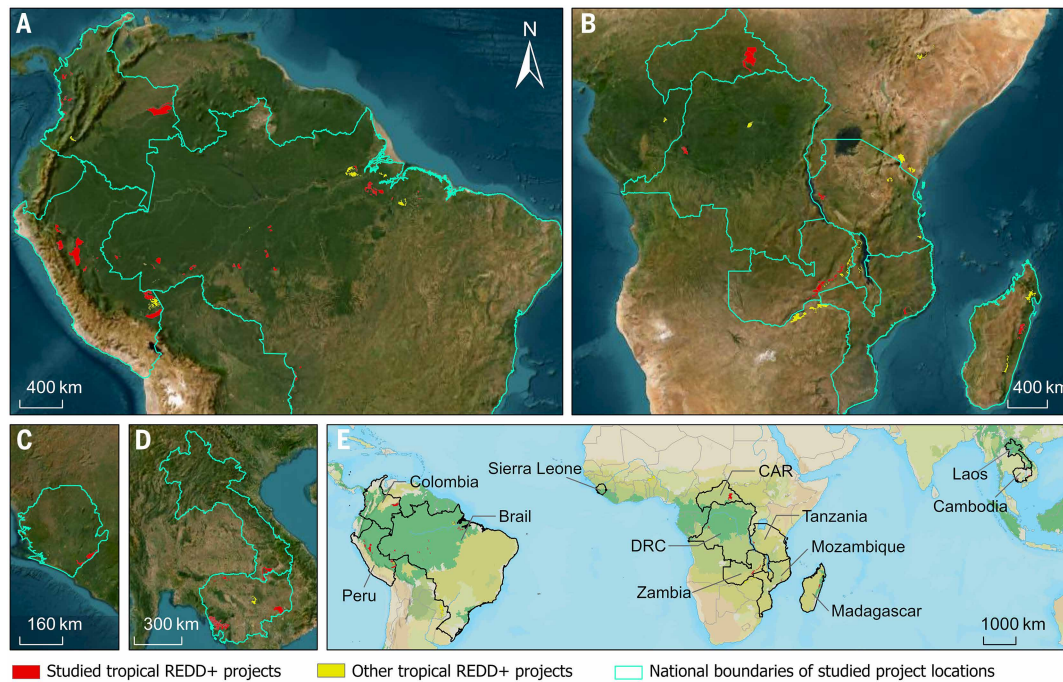


Fig. 1. Distribution map of certified pantropical REDD+ projects in the AUD category during 2001 to 2023. (A) Latin America, including Brazil, Colombia, and Peru. (B) Most of the studied countries in Africa, including Central African Republic (CAR), the Democratic Republic of Congo (DRC), Madagascar, Mozambique, Tanzania, and Zambia. (C) Sierra Leone. (D) Southeast Asia, including Cambodia and Laos. (E) Global view. The base map data is from Esri.

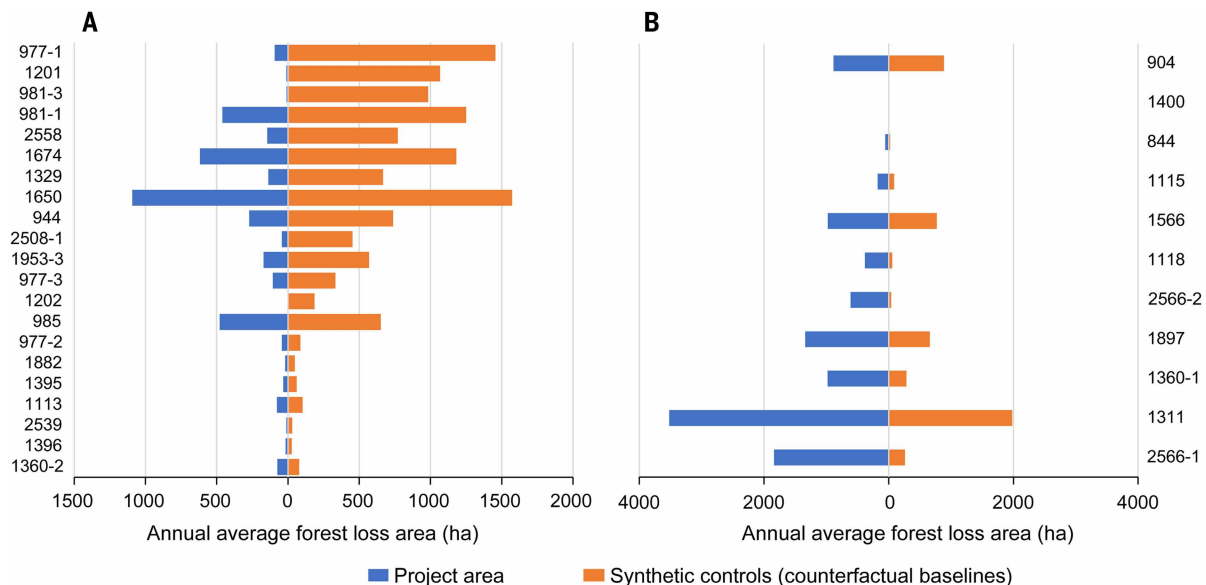


Fig. 2. Post-REDD+ annual average forest loss in project areas versus that of their synthetic controls (counterfactual baselines), shown for the 32 project units with statistically significant differences. (A) Effective projects with significant forest loss reduction (climate benefit). (B) Ineffective projects with significant forest loss increase (no climate benefit). Results for 34 of 66 total project units with insignificant differences are not shown. For project ID descriptions and geographical locations, see table S1 and fig. S3.

REDD+ project effectiveness

Nearly one-third of the project units (21/66) experienced significantly lower forest loss than expected, suggesting likely successful climate mitigation (Fig. 2A, figs. S7 and S8, and table S5). Of these, one unit of the Portel-Pará REDD+ project (977-1) in the Brazilian Amazon had the greatest reduction in forest loss area compared with that of its control (a decrease of 1362.5 ha year⁻¹ after REDD+ implementation, totaling 19.1 thousand ha saved by 2022). Likewise, a unit in the Pacajai

REDD+ Project (981-3), also in the Brazilian Amazon, had the largest percentage decrease of forest loss (a 99% reduction). However, 17% (11/66) of the project units actually had higher losses of forests designated in the REDD+ area compared to that of its control (Fig. 2B). Of these, the Brazilian Serra do Amolar project (2566-1) and the Ankeniheny-Zahamena Corridor project (1311) in Madagascar both experienced the highest forest loss (loss of >1500 ha year⁻¹) in project sites compared with that of their controls.

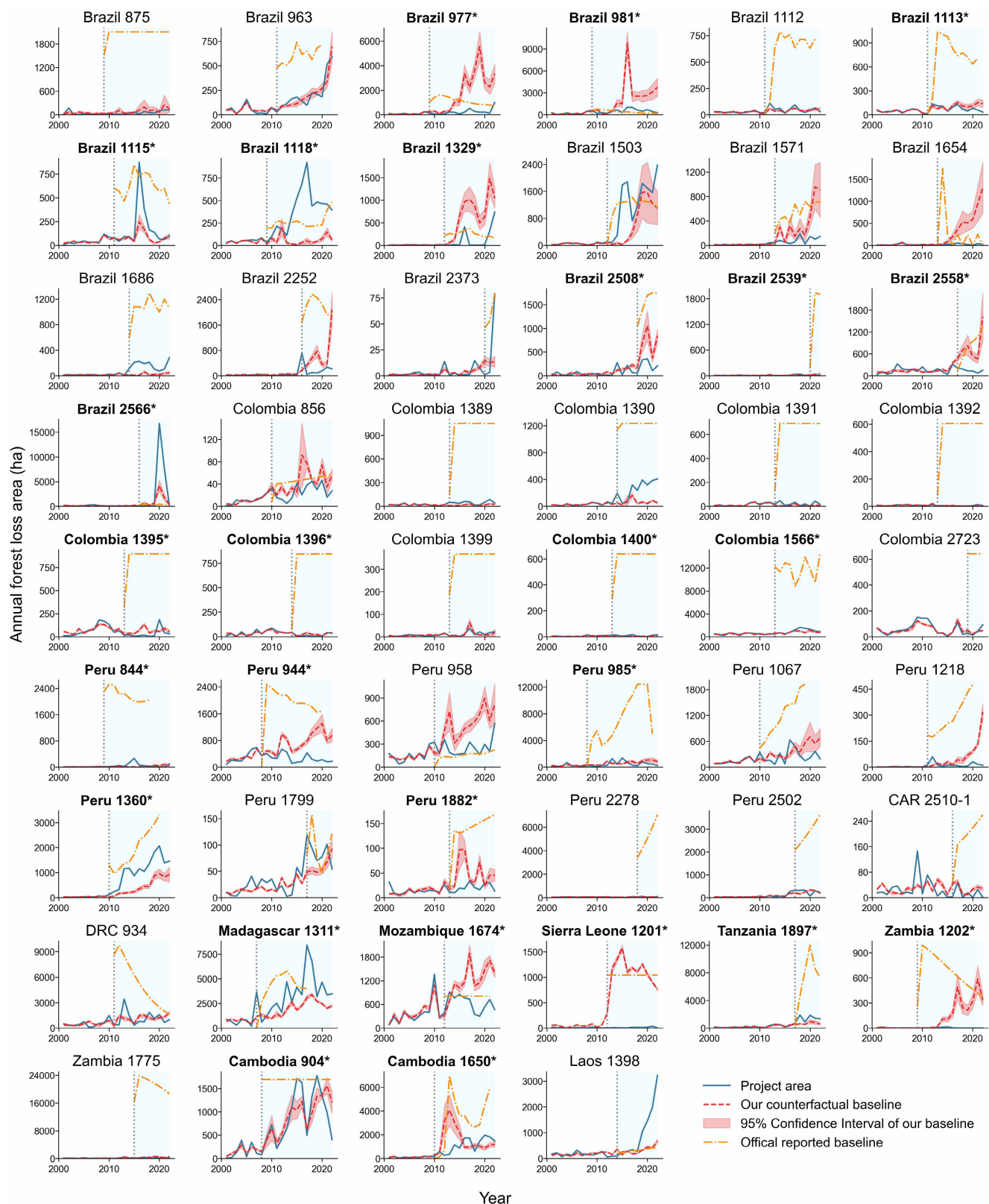


Fig. 3. Annual post-2000 forest loss in studied REDD+ project areas versus that of their synthetic controls and official reported baseline deforestation. Blue lines represent the annual observed forest loss area in project areas, whereas dashed red lines represent that in synthetic controls. Shaded red areas indicate 95% confidence intervals around our baseline deforestation averages, determined from 10 qualified synthetic control models for each individual project. Dash-dotted yellow lines demonstrate the official baseline annual deforestation curves. Dotted black lines indicate the REDD+ project implementation year, and the light blue panel demonstrates the implementation period. Bold subplot titles with asterisks indicate a significant change in deforestation in the project areas compared with that of their synthetic controls, according to the placebo tests and 95% confidence intervals. For project ID descriptions and geographical locations, see table S1 and fig. S3.

The levels of baseline deforestation risks reported in the REDD+ project description documents were substantially different than those that we observed. About 35% (18/52) of the projects had official baselines more than 10 times higher than our estimates, causing substantial overestimation of the forest loss reductions caused by REDD+ projects. Colombia is a hotspot, where 10/12 projects reported more than 10-fold-higher baselines than we estimated. By contrast, ~19% (10/52) of the projects had substantially less deforestation than expected, implying that the deforestation risk of these projects were conservatively estimated (Fig. 3, fig. S9, and table S6).

When we combined results of individual project units into a synthetic analysis based on the GSCM, we found overall decreases of deforestation rate brought by the REDD+ implementation across the whole pantropical region (−0.16%); likewise, we found rate decreases in the regions encompassing Brazil (−0.33%) and Africa (−0.19%) (Fig. 4 and table S7). However, when we used a different analytical approach (SDID), we found results that were no longer significant at the $P = 0.05$ threshold ($P = 0.09$) (table S7). In this case, we favor stronger inference from the GSCM, which offers analytical flexibility by accommodating time-varying unobserved confounders, whereas the SDID relies on an assumption of parallel trends (i.e., treated and control groups follow similar trajectories without intervention). Nevertheless, using both synthetic approaches can yield a more comprehensive understanding of the effectiveness of REDD+, which, in this case, indicates some uncertainty.

Carbon credit reassessment

To reassess the tradable carbon credits generated by REDD+ projects, we compared the amount of forest loss reduction derived from our

counterfactual baseline with the official reported baselines in 48 of the 52 projects that had publicly available data from Verra on Verified Carbon Units (VCU). In the voluntary carbon market, carbon credits are tradable certificates representing the avoidance of one metric ton of carbon dioxide emissions. These credits enable purchasers to offset their own emissions by investing in verified climate projects. Of the potential 264 million tradable carbon credits that these 48 REDD+ projects could have generated by 2022, only 73 million (28%, considering only projects with significant climate benefits) are likely to emerge from the avoided unplanned deforestation activities. We estimated that ~34.8 million carbon credits from these 48 REDD+ projects could genuinely offset emissions, representing 13.2% of the total tradable carbon credits (Fig. 5 and table S8). However, by December 2022 (when the last publicly available data were reported), these projects had issued up to 228.4 million tradable carbon credits (95% of which were sold), with 127 million (55%) used by companies or individuals to offset their greenhouse gas emissions (tables S9 and S10). This represents a 5.6-fold increase over our estimate of genuinely additional credits, highlighting a significant discrepancy that raises concerns about the integrity of the current carbon credit system.

Robustness tests

To ensure that our findings were robust to key assumptions, we conducted additional tests, varying concession data inclusion, forest loss datasets, and leakage buffer widths (supplementary materials, section S5). Incorporating concession covariates modestly reduced the proportion of effective credits but did not alter overall conclusions (fig. S10 and tables S11 and S12). We also compared results using the JRC Tropical Moist Forest dataset versus the Global Forest Change

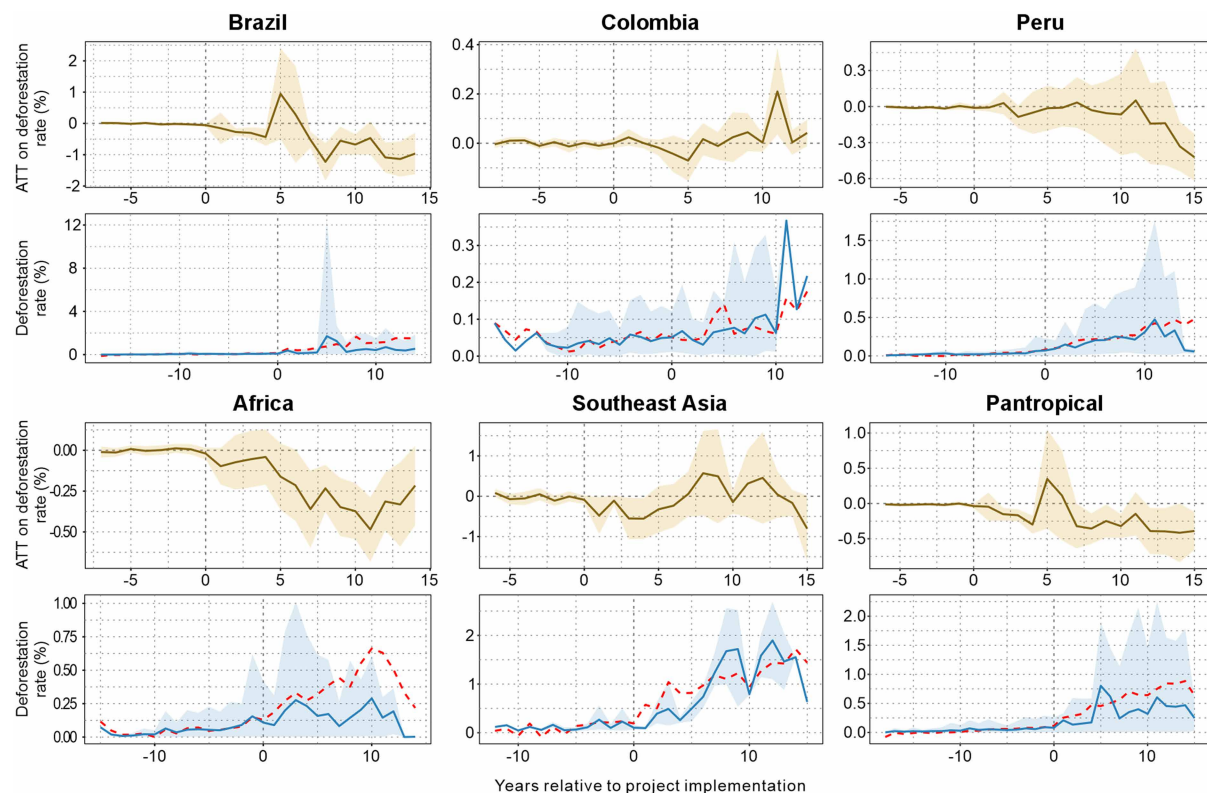


Fig. 4. Estimated average impacts of REDD+ projects on annual deforestation at the country or regional scale, including Brazil, Colombia, Peru, Africa, Southeast Asia, and the whole pantropical region, based on GSCM. (Top) The average treatment effect on treated (ATT) units on deforestation rate. **(Bottom)** Average deforestation rates for projects (solid blue line) and counterfactual baselines (dashed red line). The shaded blue areas represent the bootstrapped 95% confidence intervals around the project deforestation averages, estimated by the GSCM.

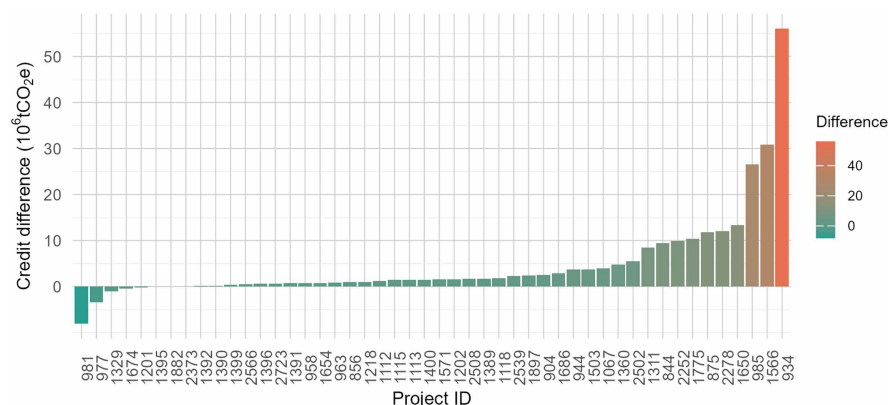


Fig. 5. Comparison of the differences in tradable carbon credits predicted in official project reports and those reassessed on the basis of our counterfactual baseline by 2022. Differences greater than zero indicate over-crediting (the claimed carbon credits exceeds the actual reductions by our counterfactual baseline) by the developers. For project ID descriptions and geographical locations, see table S1 and fig. S3.

dataset, finding highly consistent additionality assessments despite definitional differences (fig. S11 and table S13). Lastly, adjusting leakage buffer widths from 5 to 20 km produced only minor shifts in forest loss estimates (figs. S12 and S13 and table S14).

Discussion

REDD+ has played a prominent, though often contested, role in nature-based climate policy and voluntary carbon markets. Our study offers one of the most comprehensive evaluations to date, analyzing 52 voluntary REDD+ projects (66 units) across the tropics. Only a minority achieved statistically significant reductions in deforestation, and just 19% met their reported emissions targets. We evaluated these data in several ways to assess whether alternative assumptions, analyses or data sources would qualitatively alter our conclusions, finding that our results are largely robust to these changes. At the same time, many of these underperforming projects still delivered partial climate benefits, contributing to an estimated 13.2% of tradable carbon credits. As a result, despite the limited effectiveness of many ongoing projects, there is still a high potential for climate impact when REDD+ is implemented and evaluated rigorously.

Our findings that variable but generally low numbers of REDD+ sites effectively reduce forest loss, coupled with limited credible carbon offsets, is consistent with those of previous work (19, 40, 41), though we found greater mitigation potential. Specifically, by adding more robust analytical procedures and including a broader and more recent set of REDD+ projects while excluding those with incomplete data or spatial inconsistencies, we were able to estimate a nearly twofold increase in the percentage of tradable carbon credits, demonstrating genuine additionality in a larger portion of projects [i.e., 13.2% compared with the 6 to 7% reported in earlier studies (17, 42)]. This includes stronger performance in specific regions: nearly 40% of Brazilian project units and several African projects showed significant reductions (figs. S14 and S15), highlighting the potential for success when implementation is robust. Even when we compared our results with those of previous work, using only sites that are shared between studies (17, 20), we identified more than twice as many with significant positive impacts (15% versus 6%). Two methodological advances (detailed in table S15) likely explain our findings: (i) the use of ecologically matched control areas combined with annual-scale synthetic control models incorporating enhanced covariates and (ii) the exclusion of avoided degradation when reassessing carbon credits.

Despite more optimism for tropical voluntary REDD+ projects that emerges from our results, we also confirm that many projects

substantially over-credit their climate benefits. This overestimation is often linked to the selection of inappropriate reference regions that are typically large tracts of land that maintain overall similarity to the conditions in the project area but also include heavily human-impacted lands that have high historical deforestation rates (fig. S16) (17, 40, 43). To address this, we advocate for stricter limits on the size of baseline reference areas and alignment with pre-REDD+ deforestation trends. Specifically, we recommend adopting more robust counterfactual construction methods, such as the SCM, which explicitly match project and control areas based on preintervention trends while maintaining ecological and socioeconomic comparability. This approach should include generating donor pools within spatially and ecologically constrained buffers, using multi-round random sampling to reduce selection bias, and incorporating covariates that capture the key spatiotemporal characteristics of the project area. These refinements would produce more credible and manipulation-resistant

counterfactual baselines while remaining compatible with REDD+ monitoring requirements.

Another source of over-crediting lies in the assumption that historical deforestation trends can reliably forecast future risk. In reality, deforestation is influenced by shifting socioeconomic and political conditions (44, 45). In some cases, high deforestation rates can decrease with economic growth (41), whereas in other cases, the relationship between per capita income and deforestation is more complex (46). In either case, historical deforestation data can sometimes poorly forecast future trends. Political shifts can also trigger abrupt changes. For example, deforestation rates in the Brazilian Amazon more than doubled from 2019 to 2021 under President Bolsonaro (47, 48), a pattern reflected in our findings from Brazil (fig. S7). Explicitly incorporating country-specific governmental cycles into Verra's methods (49) along with integrating counterfactual-based approaches could enhance the accuracy of REDD+ forecasts.

Project-level emission-reduction estimates and forecasts remain highly uncertain, especially in tracking forest area change over time, quantifying carbon stored in forest biomass, and estimating emissions per unit of forest lost or degraded (13, 14, 50). However, these errors tend to cancel out when assessed across larger portfolios, reducing the risk of overestimating total climate benefits (51). Individual REDD+ projects may overstate their impact because of measurement imitations or even intentional manipulation, but diversified portfolios help offset this by combining projects with demonstrably reliable additionality. Given growing public concern over the use of low-cost, high-risk carbon credits by high-emitting corporations (52), we recommend applying discount factors to single-project offsets to reflect those uncertainties and prioritizing the use of diversified credit portfolios. Recent efforts to improve carbon market quality and third-party ratings are important steps toward increasing transparency, reliability, and standardization in how credits are assessed and valued (53, 54). Our findings also underscore the need for targeted reform for REDD+ projects, rather than complete rejection. Priorities include improving baseline methodologies, increasing transparency in certification, and strengthening accountability to reduce over-crediting. Focusing on areas with strong, long-term carbon storage potential paired with stronger oversight could improve both credibility and outcomes.

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SUPPLEMENTARY MATERIALS

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Editor's summary

Forest degradation is a major contributor to greenhouse gas emissions, such that forest protection projects are used as carbon offsets for greenhouse gas-producing activities in voluntary carbon markets. Recently, the effectiveness of carbon-offsetting programs such as REDD+ (Reducing Emissions from Deforestation and Degradation Plus) has been called into question, causing a decline in carbon offset values. Tang *et al.* addressed this question with updated analyses incorporating many projects and carefully chosen controls. Across 52 REDD+ Avoided Unplanned Deforestation projects from 12 countries in South America, Africa, and Southeast Asia, the authors found greater climate benefits than previous studies but low efficacy overall: Less than 20% of projects met their emissions targets. Project success and overestimation both varied considerably between countries. —Bianca Lopez

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